

Ryan Cho

Staff Software Engineer | Backend & Distributed Systems

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PROFESSIONAL SUMMARY

Backend engineer with deep experience designing distributed services, identity systems, and event-driven platforms on AWS. Led modernization of authentication and authorization across multi-service environments using TypeScript, Node.js, and NestJS. Partner with product and engineering leadership to deliver reliable APIs, mentor senior engineers, and raise engineering standards through testing, CI/CD, and observability.

WORK EXPERIENCE

Software Engineer

May/2022 - May/2026

Aspen Capital

Remote | 10th Floor, New York

- Led **backend platform modernization** by decomposing legacy monolith endpoints into NestJS microservices, enabling independent deployments across portfolio analytics, reporting, and investor-facing workflows.
- Architected an **OAuth 2.0 and OIDC identity layer** with RBAC policies governing access to sensitive financial data across internal tools and partner-integrated REST APIs.
- Designed **event-driven ingestion pipelines** using AWS EventBridge and Lambda to propagate market data updates to downstream consumer services with clear service boundaries.
- Built **TypeScript and Fastify REST APIs** for investment operations teams, applying Zod schema validation and Jest/Supertest coverage to harden contract behavior under production load.
- Migrated critical persistence workloads to **Aurora PostgreSQL and DynamoDB** patterns tuned for read-heavy portfolio queries and low-latency session metadata storage.
- Championed **CI/CD, observability, and service ownership** standards with automated integration tests and runbooks that shortened incident recovery for production backend services.
- Mentored engineers on **distributed system design**, code review quality, and maintainable Node.js patterns while scaling the platform engineering footprint.
- Partnered with product and compliance stakeholders to ship **entitlement-aware platform features** that met security, audit, and cross-team integration requirements.

Technology Analyst, Incoming Technology Analyst, Technology Summer Analyst,

May/2019 - Apr/2022

BNP Paribas

Remote | Seventh Avenue, New York

- Developed **Node.js and TypeScript backend services** supporting trading, risk, and internal operations applications within a service-oriented banking technology environment.
- Implemented **authorization middleware** enforcing ABAC rules for internal tools used by global technology and operations teams across regulated business units.
- Contributed to **API gateway integration** standardizing authentication flows between legacy mainframe-adjacent systems and cloud-hosted microservices.
- Built **asynchronous job processors** for batch reconciliation workflows, improving reliability of end-of-day financial reporting and audit trail generation.
- Designed **SQL data access layers** for high-volume transaction stores requiring strict consistency, indexing strategy, and query performance for compliance reporting.
- Participated in **cross-team architecture reviews** for distributed system changes affecting multiple business lines, documenting trade-offs and migration risks.
- Automated **CI/CD testing and deployment workflows** for regulated production releases, increasing release confidence across staging and production environments.
- Collaborated with senior engineers to define **service contracts, error handling conventions, and observability standards** adopted by multiple backend teams.

Software Engineering Intern

Jun/2018 - May/2019

Chexology

Remote | 17th Floor, New York

- Built **full-stack features** for a restaurant compliance platform using React and Node.js, delivering REST endpoints consumed by mobile inspection workflows.
- Implemented **session management and role-based access** for internal dashboards supporting field operations and quality assurance teams.
- Wrote **Jest unit and integration tests** validating API behavior and regression scenarios before production releases.
- Contributed to **Agile sprint delivery**, participating in code reviews and technical design discussions for new service endpoints.
- Refactored legacy Express routes into **modular service handlers**, improving maintainability as the engineering team expanded feature scope.

Web Developer

Sep/2017 - Nov/2017

Bayesquare Foundation

Remote | New York

- Developed **responsive web interfaces and backend endpoints** for a nonprofit fundraising platform serving regional community programs.
- Integrated **third-party authentication** for donor accounts and administrative users across staging and production environments.
- Optimized **PostgreSQL queries** backing reporting views used by foundation stakeholders during active campaign cycles.
- Collaborated with remote teammates on **scripted deployment workflows** to ship incremental platform updates with minimal downtime.
- Documented **REST API usage patterns** for volunteer developers extending platform functionality and content management features.

EDUCATION

New York University

Bachelor of Arts (B.A.), Computer Science

2016-2020

SKILLS

Frontend: React, TypeScript, HTML/CSS, Responsive UI

Backend: Node.js, NestJS, Fastify, TypeScript, REST APIs, Event-Driven Architecture, OAuth 2.0, OIDC, RBAC/ABAC

Databases: PostgreSQL, Aurora, DynamoDB, SQL, NoSQL, Query Optimization

Cloud & DevOps: AWS (Lambda, EventBridge, S3, ECS), Docker, Kubernetes, CI/CD, GitHub Actions, Serverless

Monitoring & Tools: Jest, Vitest, Supertest, Zod, Datadog, CloudWatch, AI-Assisted Development

Workflow & PM: Agile/Scrum, Cross-Functional Leadership, Technical Mentorship, Code Review, Service Ownership